Partial Literature Answer for a Basis-Reliance Question in arXiv:1504.06862

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Status

This note records a partial later-literature answer to the basis-reliance remark after Theorem 1.2 of Kurka, arXiv:1504.06862. The later paper is Kurka, arXiv:1507.03899.

The answer is partial: it removes the input monotone-basis hypothesis for analytic families of spaces with separable dual and for analytic families of separable reflexive spaces. It does not settle every class appearing in the source theorem, and it does not address the separate Schur-space issue from Remark 2.4(c).

Source Signal

The source paper proves an isometric amalgamation theorem for analytic classes whose members have monotone bases. Immediately after the theorem, it says that the author does not know whether the reliance on a basis can be dropped, as in the isomorphic setting.

The theorem includes, among other classes, spaces with a monotone shrinking basis and reflexive spaces with a monotone basis.

Later Partial Answer

The later paper arXiv:1507.03899 explicitly returns to this obstruction. Its introduction says that the technique from arXiv:1504.06862 applies only to spaces with a monotone basis, so the basis-reliance problem appears again, and that for two considered classes the problem can be solved by the same method as in the isomorphic setting.

Its main theorem states:

1. If $\mathcal{C}$ is an analytic set of Banach spaces with separable dual, then there is a Banach space E with a shrinking monotone basis which contains an isometric copy of every member of $\mathcal{C}$.

2. If $\mathcal{C}$ is an analytic set of separable reflexive Banach spaces, then there is a reflexive Banach space E with a monotone basis which contains an isometric copy of every member of $\mathcal{C}$.

Consequently, in these two branches the input-class basis hypothesis from arXiv:1504.06862 is no longer needed. The later theorem gives isometric copies; this note does not claim that the copies are 1-complemented.

Scope

This packet is a literature-status record, not a new theorem. It should be indexed as a partial subcase answer. It does not resolve the source theorem's non-isometrically-universal class without bases, the strictly convex class without bases, or the Schur-tree question from Remark 2.4(c).

References

1. O. Kurka, *Amalgamations of classes of Banach spaces with a monotone basis*, arXiv:1504.06862.
2. O. Kurka, *Zippin's embedding theorem and amalgamations of classes of Banach spaces*, arXiv:1507.03899.